Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 39)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why a moderate water shortage reduces crop productivity even though the plants survive.

[3 marks]

0 1 . 2 Explain the problems caused by waterlogged soils and describe **two** ways farmers reduce soil water levels.

[4 marks]

0 1 . 3 Explain how irrigating with groundwater can cause soil salinisation and how this harms crops.

[2 marks]

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0 1 . 4 State how a farmer could raise the pH of an acidic soil and explain why a low pH reduces fertility.

[1 mark]

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0 2 Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 2 . 1 Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 2 . 2 Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 2 . 3 Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

0 2 . 4 Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

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0 3 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

0 3 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

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0 3 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

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0 3 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 4 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

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Regulation of atmospheric composition .....

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Soil maintenance .....

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Interspecies relationships .....

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0 4 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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0 4 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

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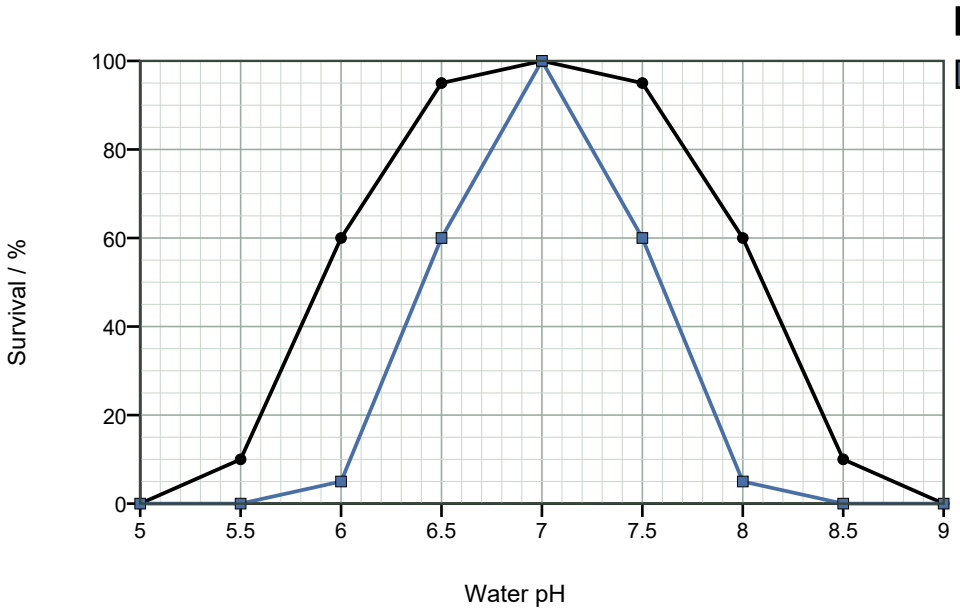
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0 5 Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 5 . 1 Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

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0 5 . 2 A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

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0 5 . 3 Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

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0 5 . 4 Give **two** biotic factors that affect the distribution of a species.

[2 marks]

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0 6 . 1 Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

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0 6 . 2 Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

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0 6 . 3 Describe **three** threats to UK broadleaf woodland.

[3 marks]

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0 7 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

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0 7 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

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0 7 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores .....

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Gas analysis of ice cores .....

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Radiocarbon dating .....

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Magnesium : calcium ratio in calcite .....

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0 8 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 8 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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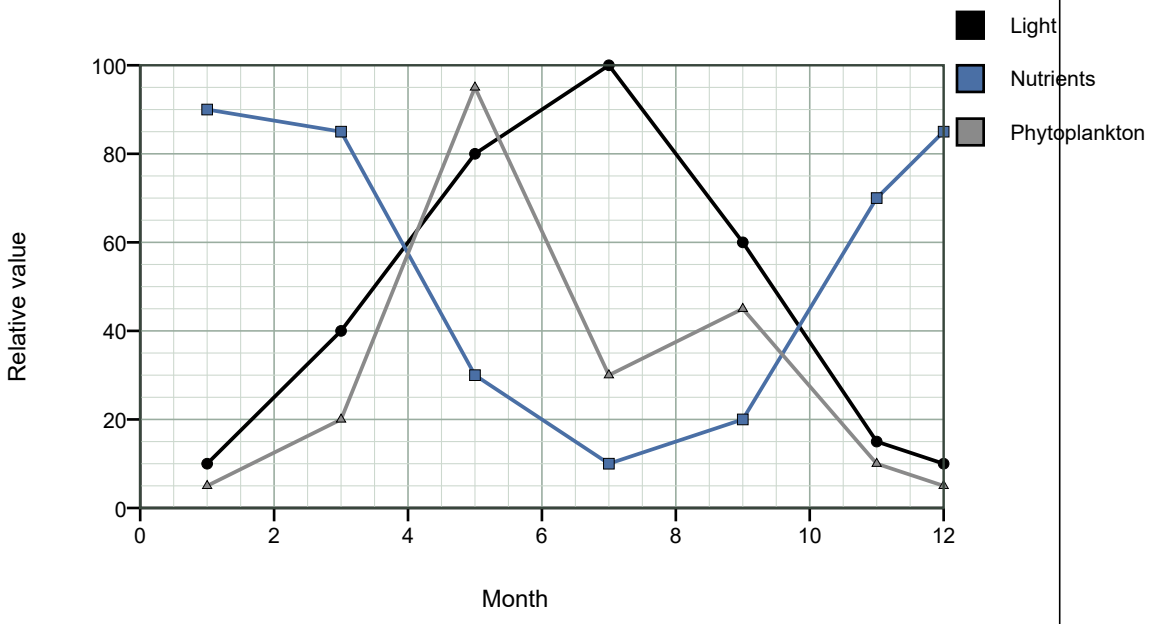
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0 9 Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 9 . 1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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0 9 . 2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

0 9 . 3 Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 9 . 4 Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

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1 0 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Extra space

1 0 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS